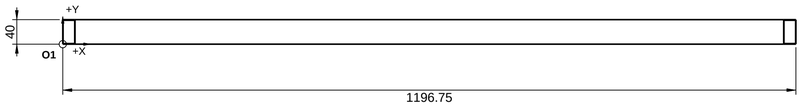


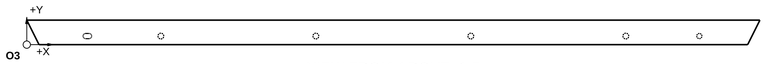
VIEW FROM BELOW 1:5

X = +X, Y = -Y (pod axes); looking up (toward +Z); origin O2



PRINCIPAL VIEW 1:5

X = +X, Y = +Z (pod axes); looking toward +Y; origin O1



VIEW FROM ABOVE 1:5

X = +X, Y = +Y (pod axes); looking down (toward -Z); origin O3

HOLES H01-H06: C1 BACK MEMBER 40X40, bottom face
seen in the view from below; origin O2; +X right, +Y up; drilled from this face

TAG	X (mm)	Y (mm)	SIZE	DEPTH	FEATURE (register) AND FITS
H01	98.88	-14.00	SLOT 8.5 x 14 OVERALL, R4.25 ENDS along X	THRU 1 WALL (1)	F-4111-10 Spigot hole and slot
H02	218.88	-14.00	Ø9.53	THRU 1 WALL (1)	F-4111-09 Back rivnut holes, J-CEIL-01
H03	471.88	-14.00	Ø9.53	THRU 1 WALL (1)	F-4111-09 Back rivnut holes, J-CEIL-01
H04	724.88	-14.00	Ø9.53	THRU 1 WALL (1)	F-4111-09 Back rivnut holes, J-CEIL-01
H05	977.87	-14.00	Ø9.53	THRU 1 WALL (1)	F-4111-09 Back rivnut holes, J-CEIL-01
H06	1097.87	-14.00	Ø8.5	THRU 1 WALL (1)	F-4111-10 Spigot hole and slot

PROJECT: **Spaceship (FocusPod v2)**	STATUS: **DRAFT**	DATE: 2026-11-01	SHEET: **1/2**

TITLE
CEILING - FRAME BACK MEMBER 40 X 40 - C1
Trần trái - thanh sau 40x40 của khung

PART NUMBER	REV	DRAWING NUMBER (FILE)
412104	**P01**	**412104_P01_CEILING-FRAME-BACK-MEMBER-40-X-40-C1_DWG**

MATERIAL
MATERIAL: Galvanized steel S280GD+Z or SGC400 (yield 280 MPa or more), Z100

FINISH
FINISH: pre-galvanized Z100; each weld's burnt zone wire-brushed and coated with 2-component zinc-rich epoxy (J-FRAME-01); the bond prep on the skin face (D-137)

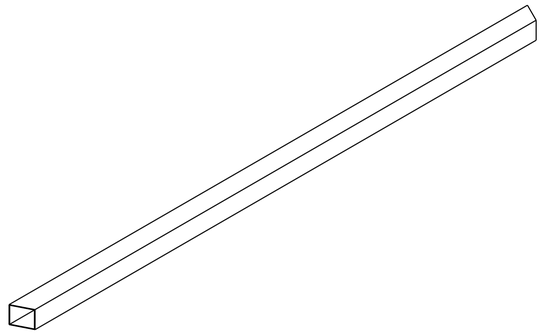
SCALE	SIZE	UNITS	MASS
1:5	A1	**UNITS: mm**	1.438 kg (calculation)

CREATED BY	APPROVED	TOLERANCES
Autonomous	not released	TOLERANCE: see the notes; not ISO 2768

PROJECTION
ISO first angle (ISO 5456-2)

GENERAL NOTES

1. UNITS: mm. Do not scale the drawing. ISO first-angle projection (ISO 5456-2), symbol in the title block. Dimensions to ISO 129-1. Sheet 1: views, sections and hole tables; sheet 2: these general notes.
2. Views: the principal view looks toward +Y (X = +X, Y = +Z (pod axes); looking toward +Y); every view's title gives its own X and Y and how they map to the pod axes (conventions section 4). On every view X runs right and Y up.
3. Origin O = the part's corner shown; O1, O2... are O seen in each view. Hole positions in the tables are from O.
4. Cut from pre-galvanized steel tube (S280GD+Z or SGC400, Z100) to the length and end cuts shown, as the frame's cut list row gives them (D-120, D-131); each mitred end cut back square to its mitre for the 2.0 mitre plate (J-FRAME-01). Welded into its frame (411101 or 411102) in the frame's jig.
5. Holes, where the tables show them: made after welding and the zinc-rich touch-up, by CNC or a drill jig that locates on the frame's joint faces and its spigot hole (J-FRAME-01); this sheet shows them on the tube for reference. Rivnuts 838003 are set after that.
6. Holes go THRU 1 WALL of the member unless the table says otherwise; deburr both sides.
7. TOLERANCES (PROPOSED until the shop confirms; not ISO 2768): cut length +0/-0.5 and end angles ±0.25° (the cut list, assumption); hole positions to the frame's datums (R-J6: Ø0.4 to A|B|C, the spigot hole and slot ±0.1); rivnut holes Ø9.53 +0.10/0 (maker); other hole sizes +0.2/0 (assumption). Not ISO 2768 on the tube.
8. Source: v2/03-Manufacturing/ceiling/410000_P01_CEILING_STEP.step (sha256 c9d36e8963f4; declared unit cm, read in mm and checked against tools/checks/stepfile.py, worst box difference 0 mm); parts list v2/03-Manufacturing/ceiling/410000_P01_CEILING-PARTS-LIST_BOM.csv; feature rationale v2/02-Drawings/ceiling/FEATURE-RATIONALE.md. From that STEP: regenerate it, never edit the PDF.



ISOMETRIC VIEW 1:5
for identification only

PROJECT: **Spaceship (FocusPod v2)**	STATUS: **DRAFT**	DATE: 2026-11-01	SHEET: **2/2**

TITLE
CEILING - FRAME BACK MEMBER 40 X 40 - C1
Trần trái - thanh sau 40x40 của khung

PART NUMBER: **412104**	REV: **P01**	DRAWING NUMBER (FILE): **412104_P01_CEILING-FRAME-BACK-MEMBER-40-X-40-C1_DWG**

MATERIAL
MATERIAL: Galvanized steel S280GD+Z or SGC400 (yield 280 MPa or more), Z100

FINISH
FINISH: pre-galvanized Z100; each weld's burnt zone wire-brushed and coated with 2-component zinc-rich epoxy (J-FRAME-01); the bond prep on the skin face (D-137)

SCALE: 1:5	SIZE: A1	UNITS: **UNITS: mm**	MASS: 1.438 kg (calculation)
CREATED BY: Autonomous	APPROVED: not released	TOLERANCES: TOLERANCE: see the notes; not ISO 2768	

PROJECTION: ISO first angle (ISO 5456-2)